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Hammura et al.

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(54) **LIQUID EJECTION HEAD**

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 200 days.

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(22) Filed: **Mar. 27, 2023**

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(65) **Prior Publication Data**

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(57) **ABSTRACT**

(30) **Foreign Application Priority Data**

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May 19, 2022 (JP) 2022-082418

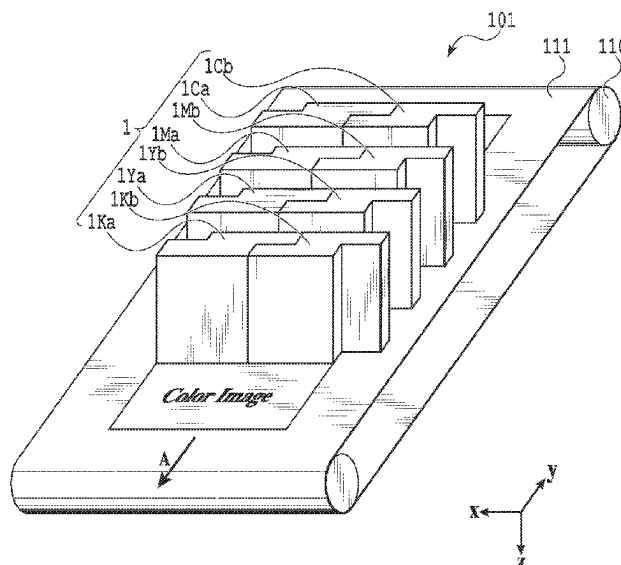
A liquid ejection head includes: an ejection port array being
an array of ejection ports; a plurality of pressure chambers
corresponding respectively to the ejection ports and com-
municating with the ejection ports; individual supply chan-
nels and individual collection channels communicating with
the pressure chambers; a common supply channel commu-
nicating with surfaces of the individual supply channels
opposite to surfaces thereof communicating with the pres-
sure chambers; a common collection channel communicat-
ing with surfaces of the individual collection channels
opposite to surfaces thereof communicating with the pres-
sure chambers; and a damper member forming a wall of a
part of at least one of the common supply channel or the
common collection channel. The common supply channel
and the common collection channel are formed so as to
extend in a first direction along the ejection port array, and
are disposed side by side in a second direction crossing the
ejection port array.

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B41J 2/14 (2006.01)

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CPC .. **B41J 2/14201** (2013.01); **B41J 2002/14419**
(2013.01)

(58) **Field of Classification Search**
CPC B41J 2/14201; B41J 2002/14419; B41J
2002/14241; B41J 2/14233; B41J
2202/12; B41J 2/04525; B41J 2/14
See application file for complete search history.

19 Claims, 12 Drawing Sheets



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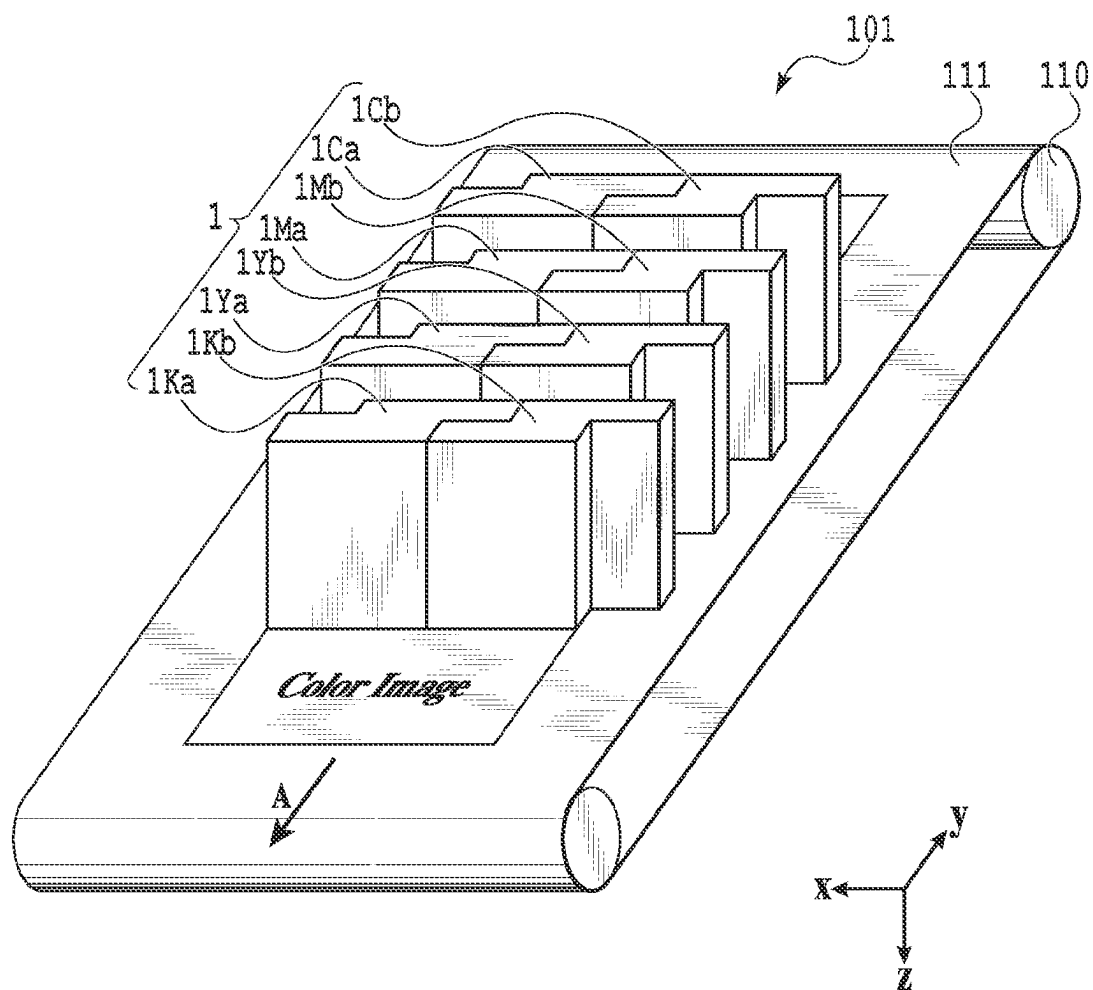
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**FIG.1**

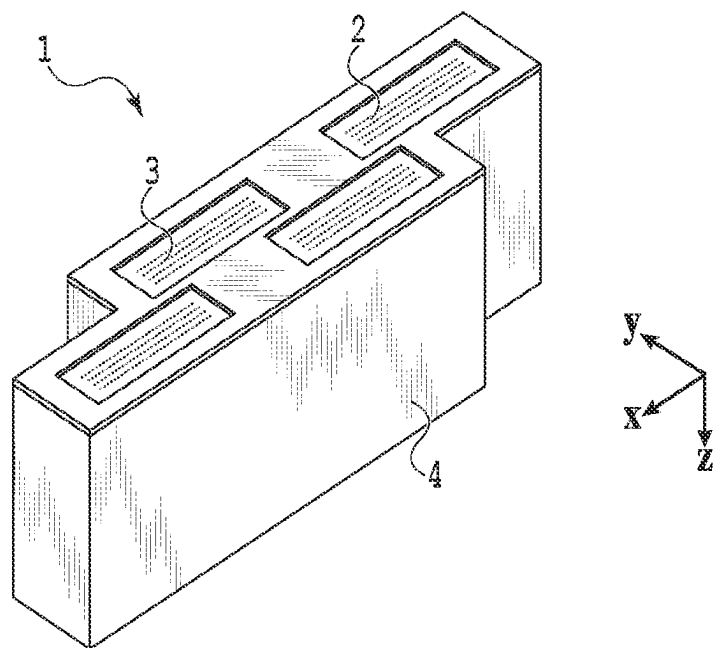


FIG. 2A

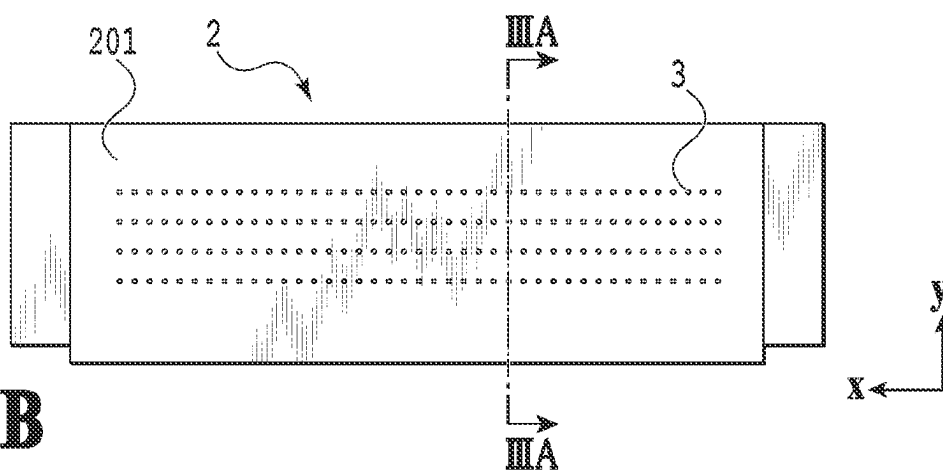


FIG. 2B

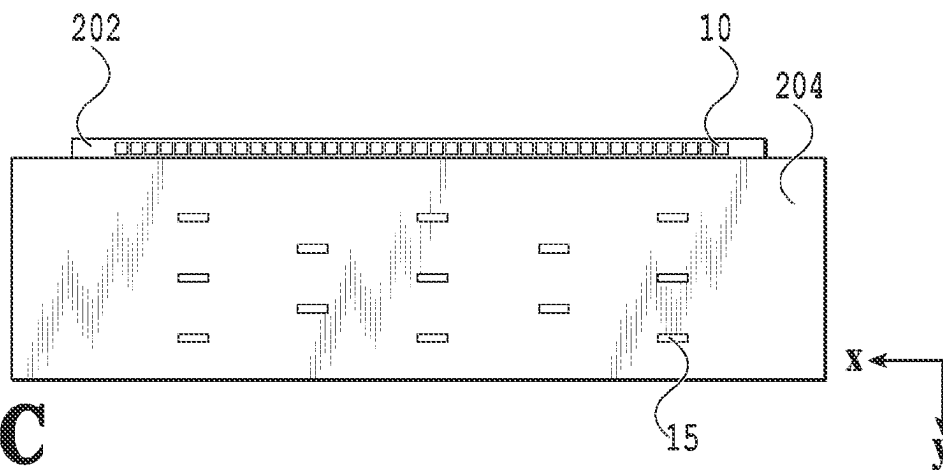


FIG. 2C

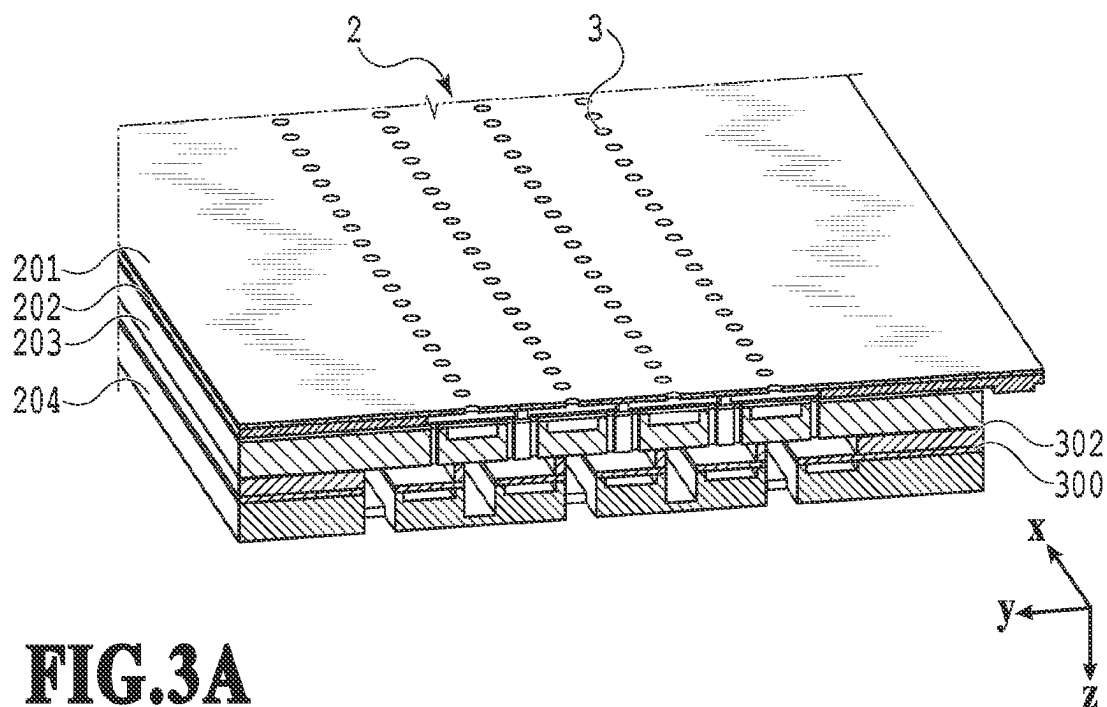


FIG. 3A

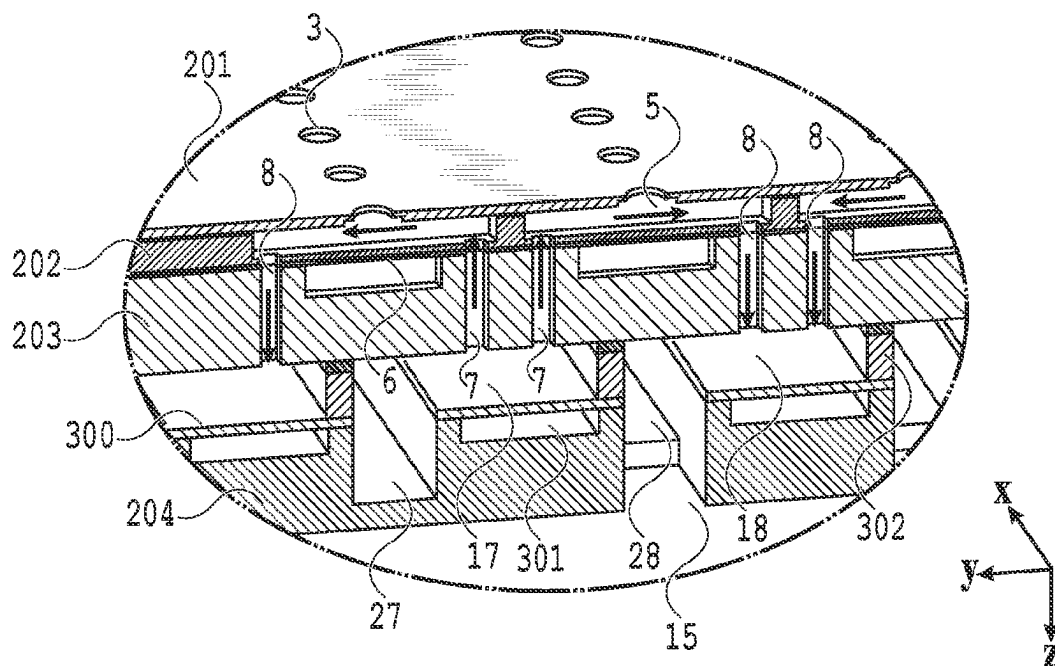


FIG. 3B

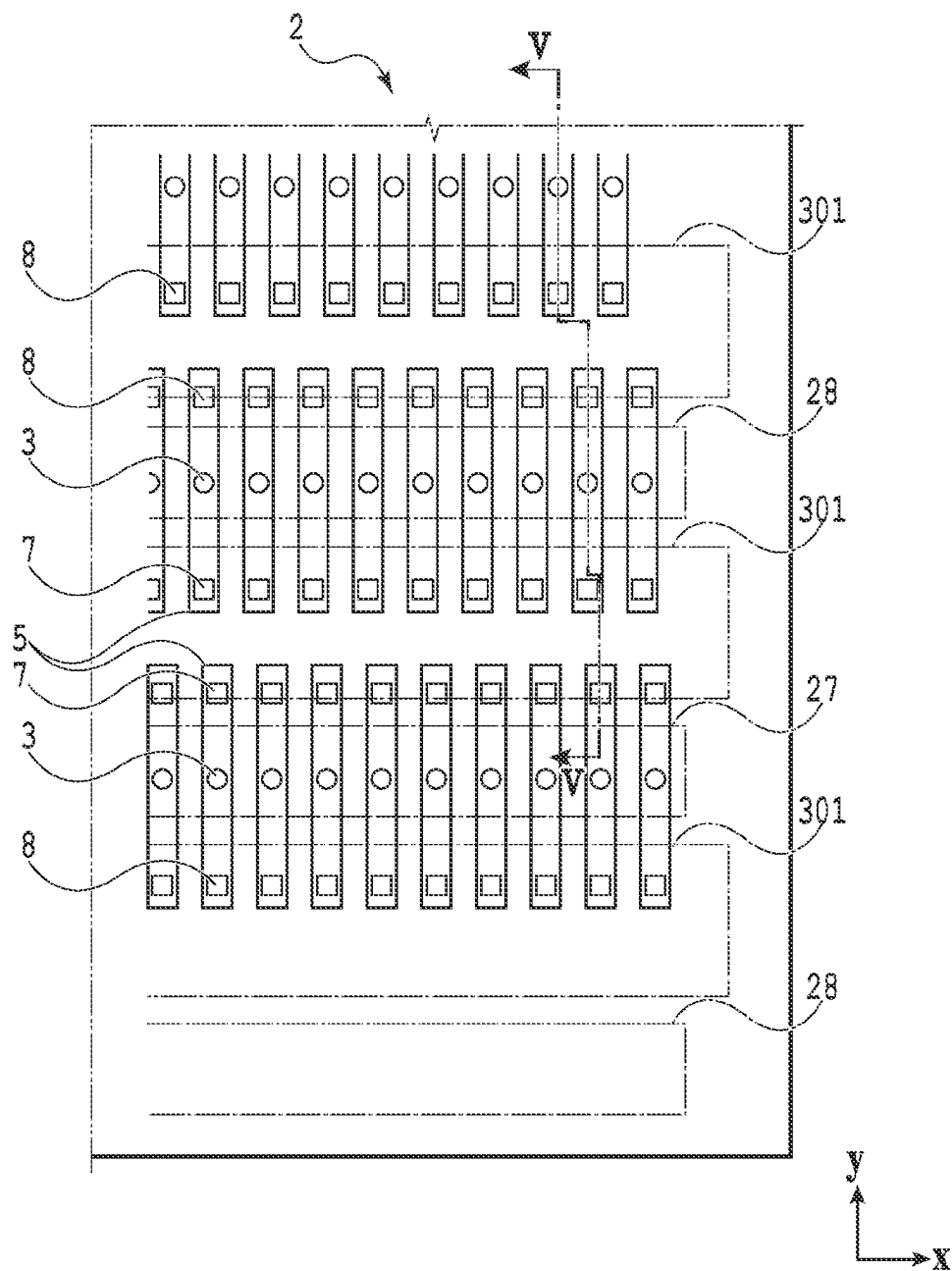


FIG. 4

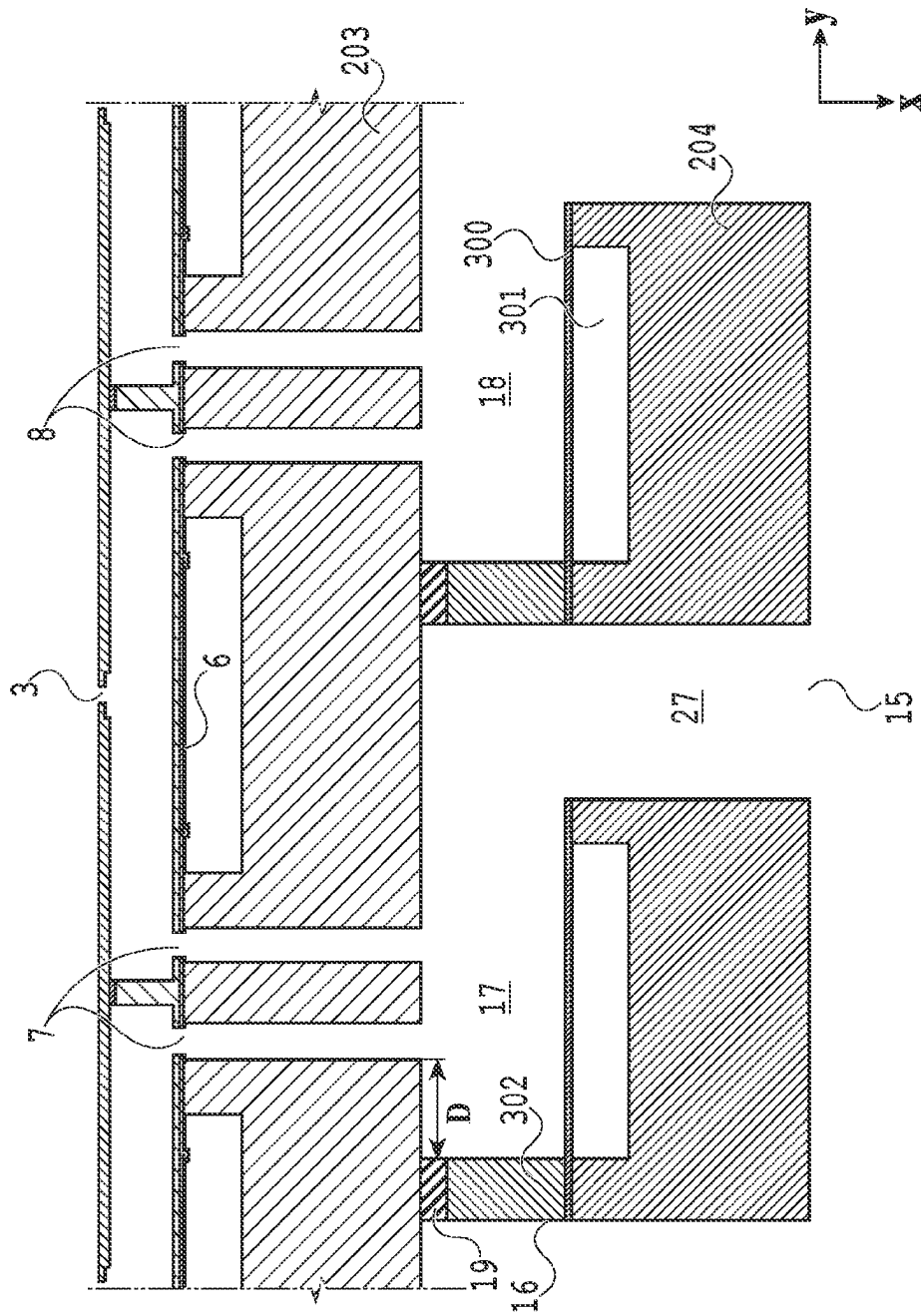


FIG 5

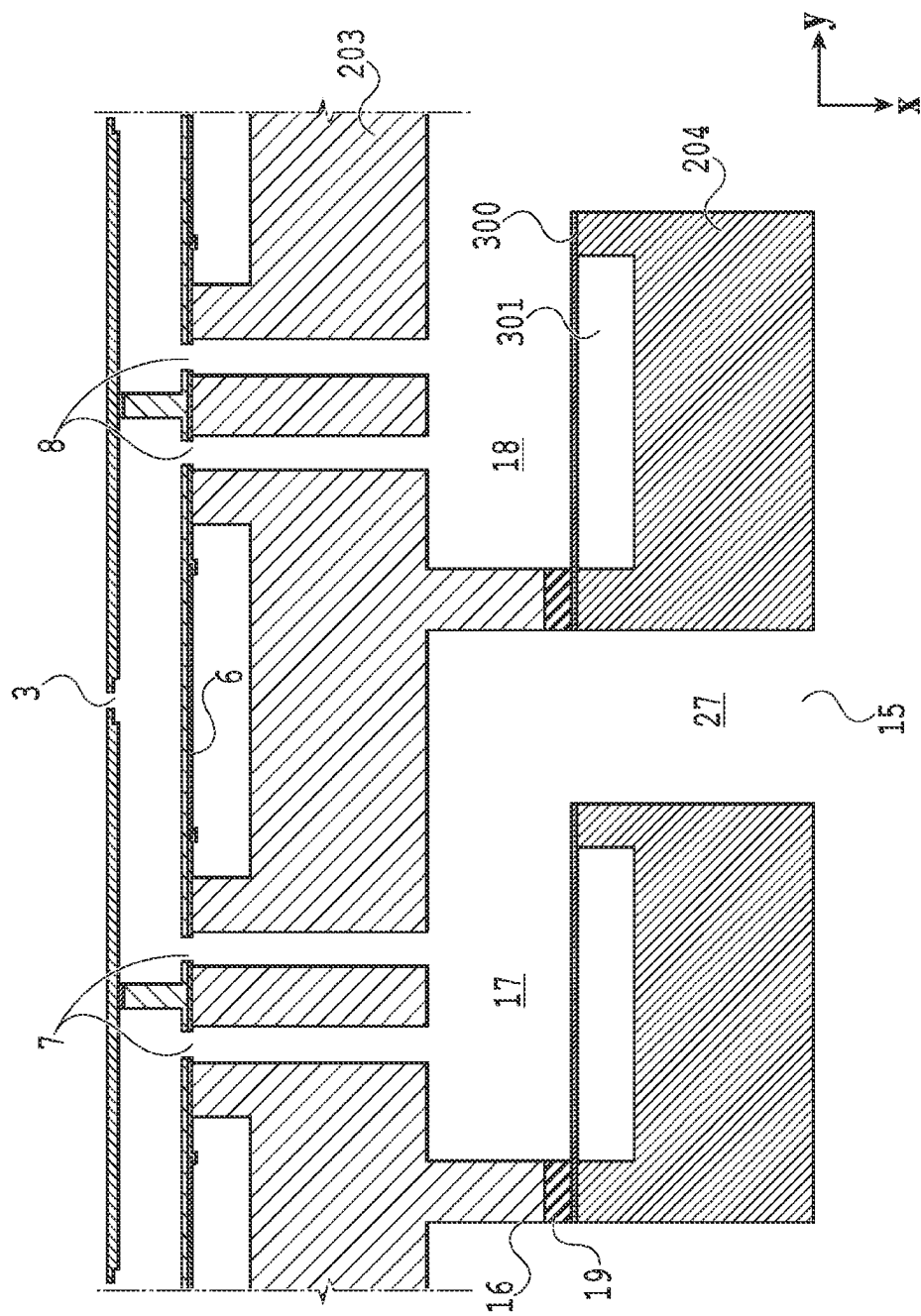


FIG. 6

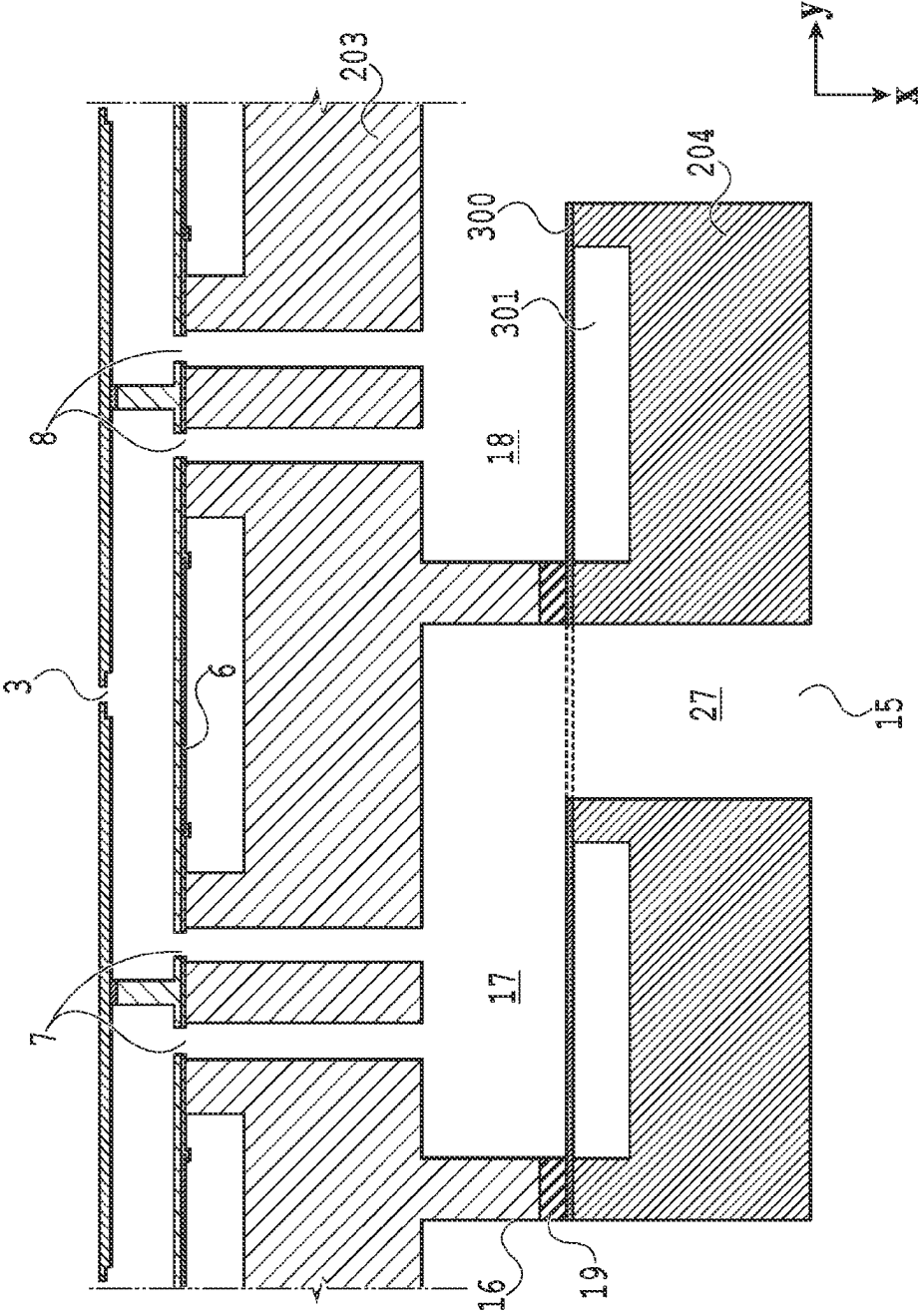
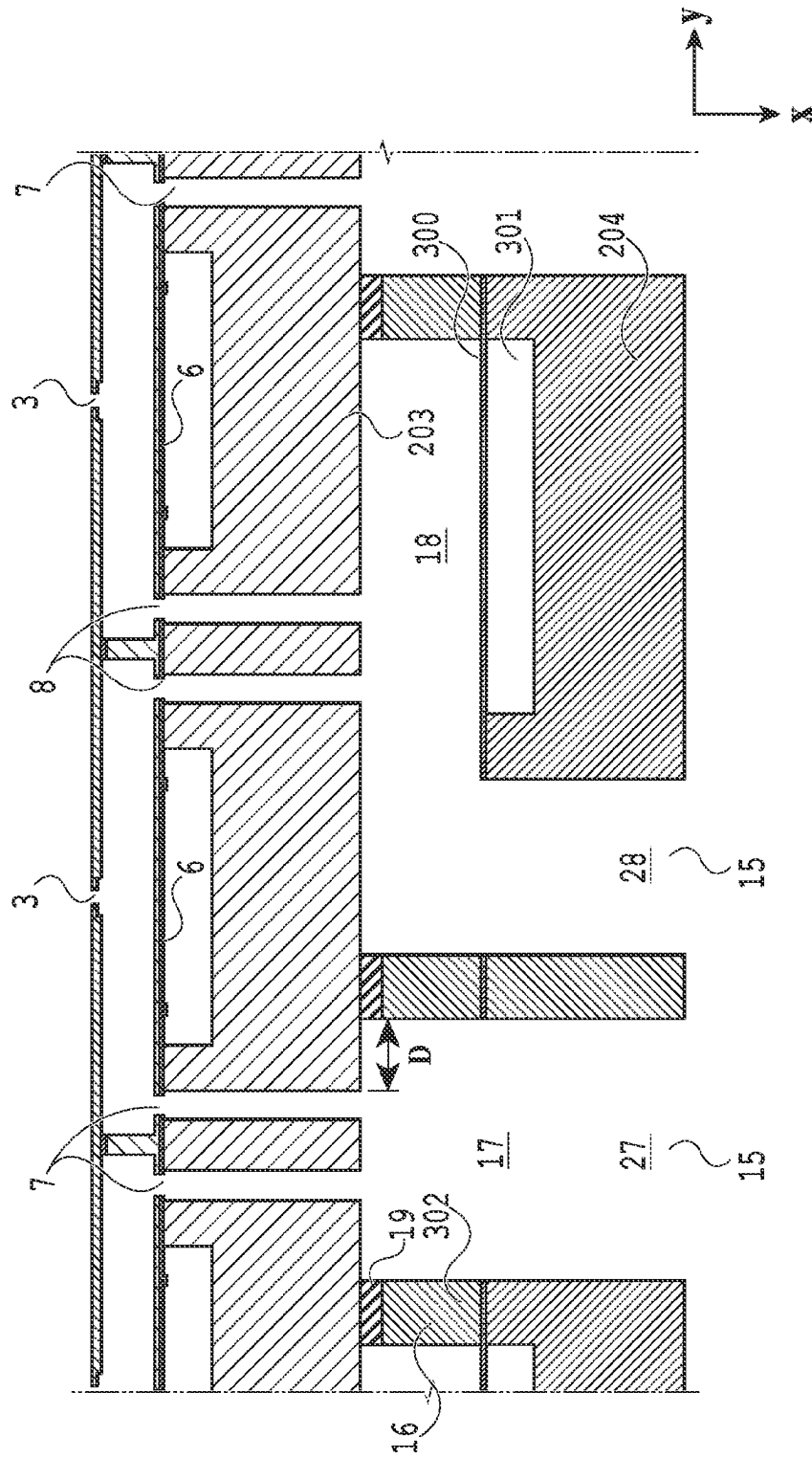
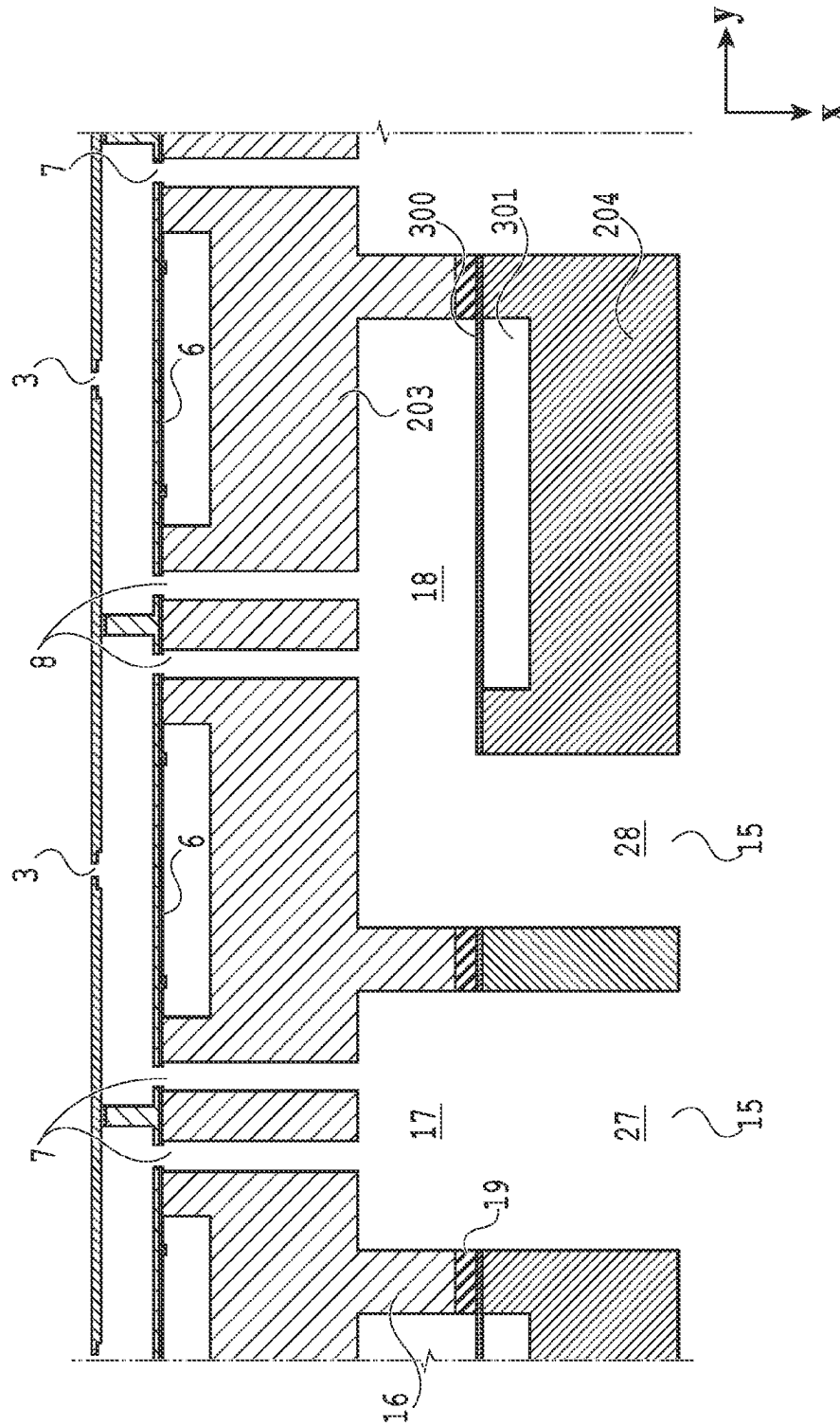


FIG. 7





9.2

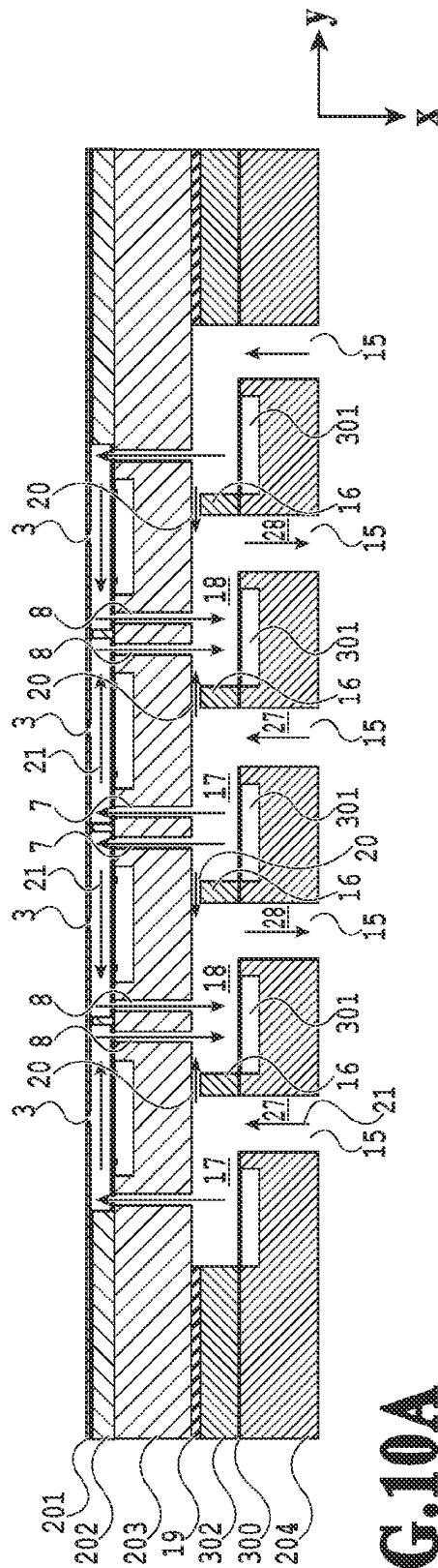


FIG. 10A

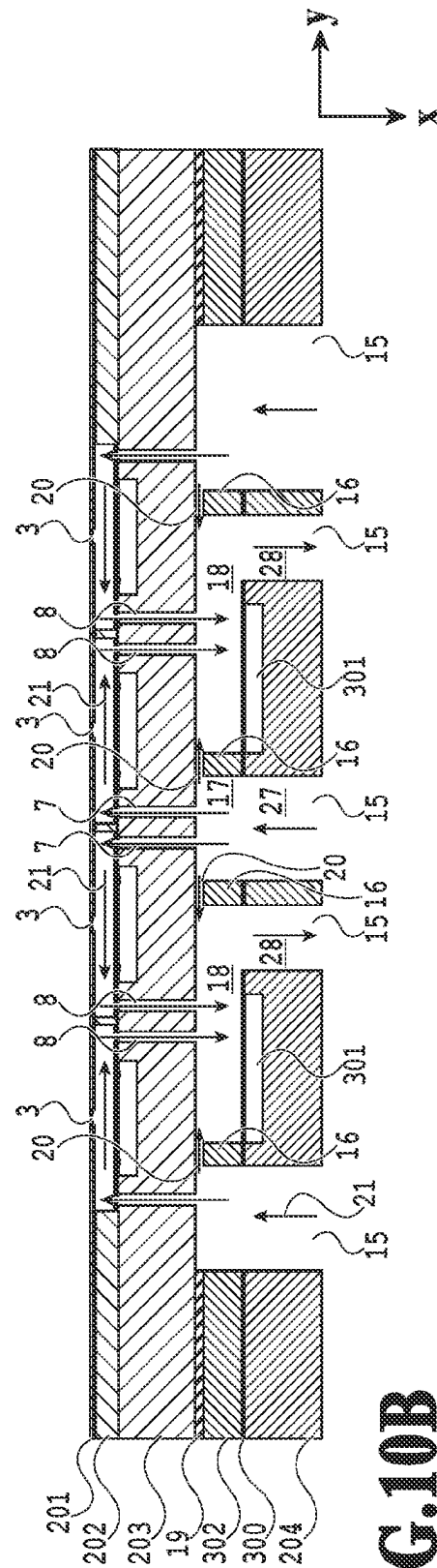
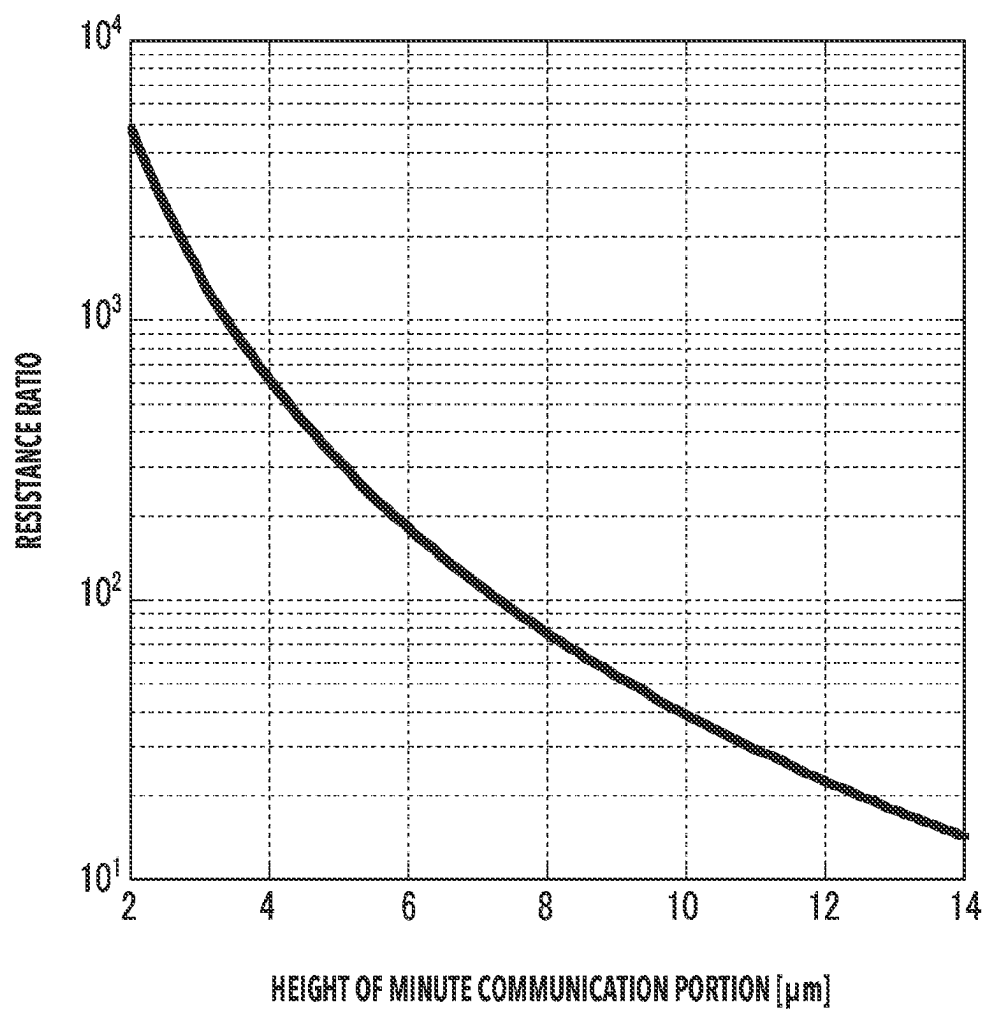


FIG. 10B

**FIG.11**

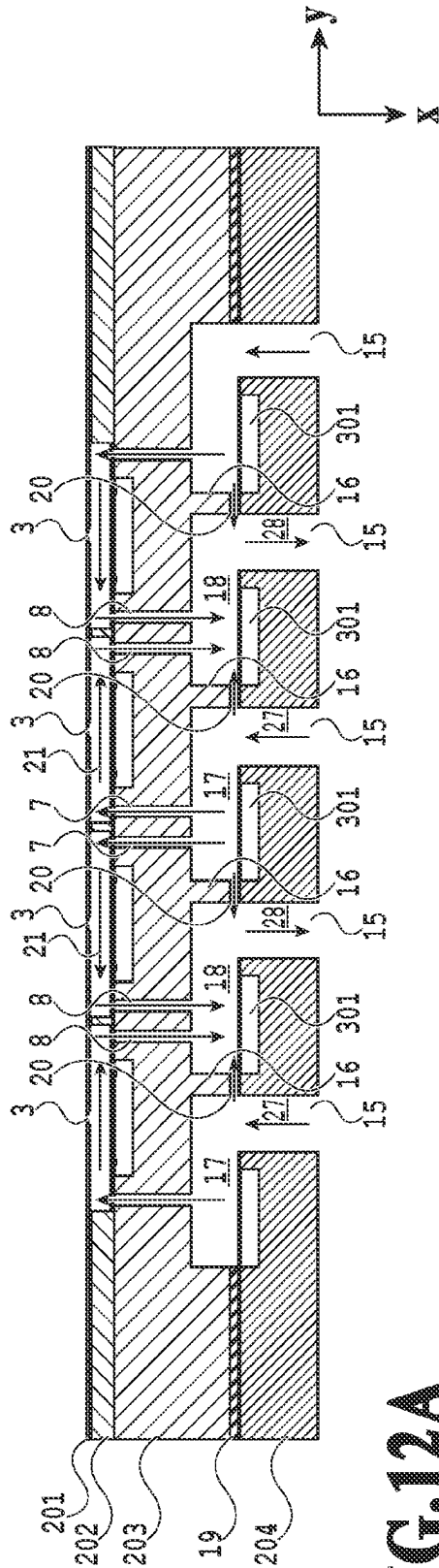


FIG. 12A

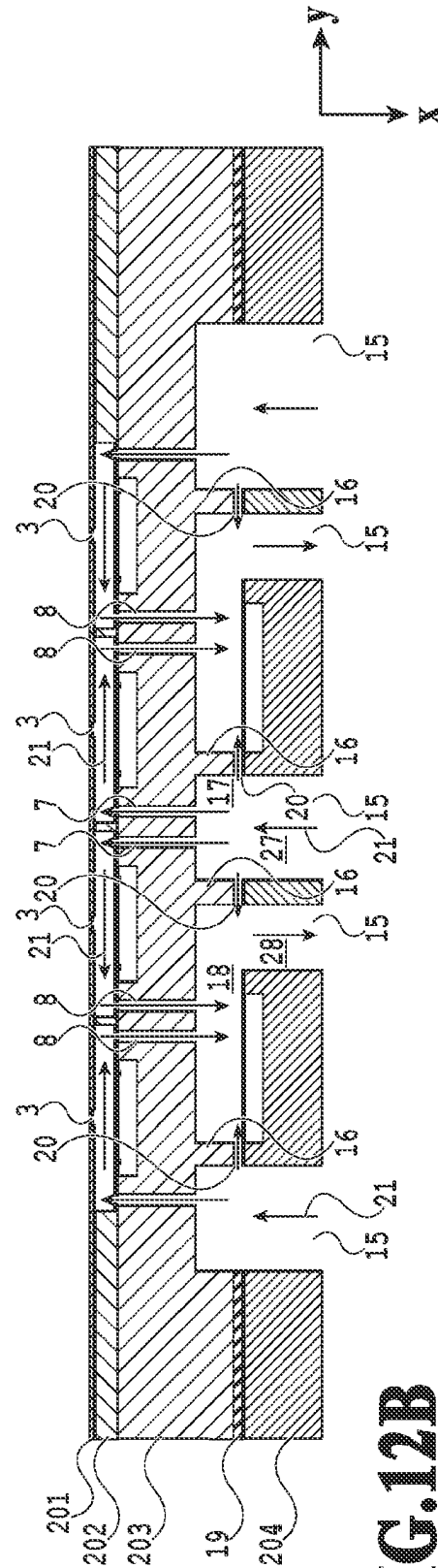


FIG. 12B

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LIQUID EJECTION HEAD**BACKGROUND OF THE INVENTION****Field of the Invention**

The present disclosure relates to a liquid ejection head.

Description of the Related Art

In liquid ejection heads that eject liquids, a phenomenon called crosstalk occurs in which a pressure fluctuation occurs in response to ejection of the liquid and this pressure fluctuation propagates to other pressure chambers through liquid channels and affects ejection characteristics. The crosstalk causes a fluctuation in ejection speed or ejection volume and may adversely affect images.

As means for suppressing such crosstalk, a configuration has been known in which liquid channels are provided with dampers to absorb pressures. To achieve a sufficient crosstalk suppression effect, the area of the dampers is required to be sufficiently wide. Incidentally, in recent years, the ejection ports in liquid ejection heads are required to be dense in order to obtain high image quality. The more densely the ejection ports are disposed, the greater the effect of the crosstalk becomes, and the wider the damper areas are required to be.

Japanese Patent Laid-Open No. 2019-155909 (hereinafter referred to as Document 1) discloses a liquid ejection head in which ejection ports are arrayed in the longitudinal direction of a substrate to thereby form ejection port arrays. Also, a rectangular pressure chamber is provided for each ejection port. For each pressure chamber, an individual supply channel and an individual collection channel are disposed. The individual supply channels and the individual collection channels communicate with branched common supply channels and branched common collection channels. In Document 1, the branched common supply channels and the branched common collection channels extend in the transverse direction of the substrate. Also, the branched common supply channels and the branched common collection channels are disposed alternately in the longitudinal direction of the substrate, in which the ejection port arrays extend. In Document 1, part of the walls of these branched channels serves as dampers and absorbs pressures from the pressure chambers to thereby suppress crosstalk.

In the configuration disclosed in Document 1, the length of the dampers is limited since they extend in the transverse direction of the substrate. This leads to a problem that a sufficient damping effect cannot be achieved and the crosstalk suppression effect is therefore low. Moreover, in Document 1, the individual channels communicate with the corresponding branched common channels, and the pressure chambers are therefore disposed with their longitudinal direction oriented in the longitudinal direction of the substrate. This makes it difficult to dispose the ejection ports such that the ejection port density is high in the longitudinal direction of the substrate.

SUMMARY OF THE INVENTION

A liquid ejection head according to an aspect of the present invention includes: an ejection port configured to eject a liquid; an ejection port array being an array of a plurality of the ejection ports; a plurality of pressure chambers corresponding respectively to the plurality of ejection ports and communicating with the ejection ports; a plurality

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of individual supply channels corresponding respectively to the plurality of pressure chambers and communicating with the pressure chambers; a plurality of individual collection channels corresponding respectively to the plurality of pressure chambers and communicating with the pressure chambers; a common supply channel communicating with the plurality of individual supply channels, the common supply channel communicating with surfaces of the individual supply channels opposite to surfaces thereof communicating with the pressure chambers; a common collection channel communicating with the plurality of individual collection channels, the common collection channel communicating with surfaces of the individual collection channels opposite to surfaces thereof communicating with the pressure chambers; and a damper member forming a wall of a part of at least one of the common supply channel or the common collection channel, in which the common supply channel and the common collection channel are formed so as to extend in a first direction along the ejection port array, and the common supply channel and the common collection channel are disposed side by side in a second direction crossing the ejection port array.

Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a view schematically illustrating a printing apparatus;

FIGS. 2A to 2C are views explaining a liquid ejection head;

FIGS. 3A and 3B are views explaining a liquid ejection substrate;

FIG. 4 is a plan view explaining channel portions in a liquid ejection substrate;

FIG. 5 is a view illustrating a cross section around ejection ports;

FIG. 6 is a view illustrating a cross section around ejection ports;

FIG. 7 is a view illustrating a cross section around ejection ports;

FIG. 8 is a view illustrating a cross section around ejection ports;

FIG. 9 is a view illustrating a cross section around ejection ports;

FIGS. 10A and 10B are views illustrating cross sections around ejection ports;

FIG. 11 is graph illustrating the height of a minute communication portion versus a resistance ratio; and

FIGS. 12A and 12B are views illustrating cross sections around ejection ports.

DESCRIPTION OF THE EMBODIMENTS

Preferred embodiments of the present disclosure will be specifically described with reference to the accompanying drawings. Note that the following embodiment does not limit the contents of the present disclosure, and not all of the combinations of the features described in the following embodiments are necessarily essential for the solution provided by the present disclosure.

First Embodiment

A liquid ejection head and liquid ejection apparatus according to a first embodiment will be described below

with reference to drawings. In the present embodiment, a liquid ejection head and inkjet printing apparatus that eject inks will be described as an example, but the present embodiment is not limited to this example. The liquid ejection head and liquid ejection apparatus according to the present disclosure are applicable to apparatuses such as printers, copiers, facsimile machines having a communication system, and word processors having a printer unit, as well as industrial printing apparatuses combining various processing apparatuses. For example, the liquid ejection head and liquid ejection apparatus according to the present disclosure are usable in applications such as fabrication of biochips and printing of electronic circuits. Also, the liquids to be ejected are not limited to inks.

<General Description of Printing Apparatus>

FIG. 1 is a view schematically illustrating a printing apparatus 101 representing an example of the liquid ejection apparatus in the present embodiment. The printing apparatus 101 in FIG. 1 has one-pass liquid ejection head modules 1 (hereinafter referred to as "liquid ejection heads 1") that print an image on a print medium 111 while moving the print medium 111 once. Ejection ports (referred to also as "nozzles") are arrayed on the sides of the liquid ejection heads 1 over the entire width of the print medium 111. The liquid ejection heads 1 in the present embodiment are heads supporting four colors of cyan (C), magenta (M), yellow (Y), and black (K). More specifically, the liquid ejection heads 1 include liquid ejection heads 1Ca and 1Cb for a cyan (C) ink and liquid ejection heads 1Ma and 1Mb for a magenta (M) ink. The liquid ejection heads 1 further include liquid ejection heads 1Ya and 1Yb for a yellow (Y) ink and liquid ejection heads 1Ka and 1Kb for a black (K) ink. The print medium 111 is conveyed in the direction of the arrow A by a conveyance unit 110, and printing is performed thereon by the liquid ejection heads 1. Note that the printing apparatus 101 illustrated in FIG. 1 is a mere example, and may be configured such that one or more liquid ejection heads 1 in any form are mountable thereon. For example, the printing apparatus 101 may have only one type of liquid ejection head or a plurality of types of liquid ejection heads other than the four types.

<Configuration of Liquid Ejection Head>

FIGS. 2A to 2C are views explaining a liquid ejection head 1 in the present embodiment. FIG. 2A is a perspective view of the liquid ejection head 1 for any one of the colors illustrated in FIG. 1. The liquid ejection head 1 has ahead main body 4. In the head main body 4, a plurality of liquid ejection substrates 2 are disposed (four liquid ejection substrates 2 are disposed in FIG. 2A). Each liquid ejection substrate 2 includes a plurality of ejection ports 3. The ink to be ejected from the liquid ejection head 1 is supplied to the liquid ejection substrates 2 from an ink tank (not illustrated) through a common supply port (not illustrated) in the head main body 4. The liquid ejection substrates 2 are disposed such that end portions of arrays of ejection ports 3 extending in an X direction overlap one another as viewed in a Y direction. Disposing the liquid ejection substrates 2 in this manner enables printing with long ejection port arrays.

FIG. 2B is a view of the liquid ejection substrate 2 as seen from the ejection ports 3 side. FIG. 2C is a view of the liquid ejection substrate 2 as seen from the opposite side to the ejection ports 3 side. The liquid ejection substrate 2 is configured of a plurality of substrates. As illustrated in FIG. 2B, the liquid ejection substrate 2 includes an ejection port formation substrate 201. The ejection ports 3 are formed in the ejection port formation substrate 201. The ejection ports 3 are disposed along the longitudinal direction (X direction,

first direction) of the liquid ejection substrate 2 (ejection port formation substrate 201) and form an ejection port array. In the ejection port formation substrate 201, a plurality of such ejection port arrays, which extend in the longitudinal direction of the substrate, are disposed side by side in a direction crossing the direction along the ejection port arrays, i.e., the transverse direction of the substrate (Y direction, second direction). As illustrated in FIG. 2C, a channel formation substrate 204 is provided on the side of the liquid ejection substrate 2 opposite to its side where the ejection ports 3 are formed. A plurality of connection channels 15 are formed in the channel formation substrate 204. Each liquid ejection head 1 in the present embodiment is configured to circulate the ink therethrough. The ink is supplied to and collected from the liquid ejection substrate 2 through the connection channels 15 formed in the channel formation substrate 204. The ink supplied to the liquid ejection substrate 2 passes through channels inside the substrates and is ejected from the ejection ports 3 and applied to the print medium 111. In the head main body 4, there is disposed an electric substrate (not illustrated) for supplying electric power and signals necessary for ejection from the ejection ports 3. This electric substrate is connected to terminals 10 on each liquid ejection substrate 2 by wirings (not illustrated). Note that the example explained in FIGS. 2A to 2C is also a mere example of the present embodiment, and the liquid ejection head 1 can be configured in any form.

<Configuration of Liquid Ejection Substrate>

FIGS. 3A and 3B are views explaining a liquid ejection substrate 2 in the present embodiment. FIG. 3A is a view illustrating a cross-sectional view along the IIIA-III A line in FIG. 2B. FIG. 3B is an enlarged view of some ejection ports in FIG. 3A and their surroundings.

As illustrated in FIG. 3A, each liquid ejection substrate 2 in the present embodiment is formed as a laminate structure of a plurality of substrates. Specifically, the liquid ejection substrate 2 has five substrates—the ejection port formation substrate 201, a vibration substrate 202, a liquid supply substrate 203, the channel formation substrate 204, and a damper substrate 302. The liquid ejection substrate 2 is formed by affixing the damper substrate 302 having a damper member 300 between the channel formation substrate 204 and the liquid supply substrate 203.

A more specific description will be given using FIG. 3B. Pressure chambers 5 communicating with the ejection ports 3 are formed in the liquid ejection substrate 2. A pressure chamber 5 is formed for each ejection port 3. Moreover, a piezoelectric element 6 is provided on a deformable wall of each pressure chamber 5 formed by the vibration substrate 202. By deforming the vibration substrate 202, the piezoelectric element 6 can pressurize the liquid in the pressure chambers 5 and eject the ink from the ejection ports 3.

In the liquid supply substrate 203, individual supply channels 7 and individual collection channels 8 communicating with the pressure chambers 5 are formed respectively for the pressure chambers 5. The ink is supplied from the individual supply channels 7 into the pressure chambers 5 and ejected from the ejection ports 3. Part of the ink can flow into the individual collection channels 8 from the pressure chambers 5. The plurality of individual supply channels 7 each communicate with a first common supply channel 17 formed in the damper substrate 302. The plurality of individual collection channels 8 each communicate with a first common collection channel 18 formed in the damper substrate 302. The wall of the first common supply channel 17 facing the individual supply channels 7 is formed by the damper member 300. Damper areas 301 are provided at

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positions opposed to the individual supply channels 7. The wall of the first common collection channel 18 facing the individual collection channels 8 is formed by the damper member 300. Damper areas 301 are provided at positions opposed to the individual collection channels 8. The damper areas 301 are areas by the walls where the damper member 300 is formed, and are areas forming recessed spaces in the channel formation substrate 204. In a case where a pressure fluctuation occurs, the damper member 300 can absorb the pressure by using the recessed spaces provided in the channel formation substrate 204. The first common supply channel 17 and the first common collection channel 18 extend in the longitudinal direction of the liquid ejection substrate 2. Also, a plurality of first common supply channels 17 and a plurality of first common collection channels 18 are formed alternately in the transverse direction of the liquid ejection substrate 2.

The first common supply channels 17 each communicate with a second common supply channel 27 formed in the channel formation substrate 204. A plurality of connection channels 15 are formed in the second common supply channel 27. The ink is supplied from the outside of the liquid ejection substrate 2 through these connection channels 15. The first common collection channels 18 each communicate with a second common collection channel 28 formed in the channel formation substrate 204. A plurality of connection channels 15 are formed in the second common collection channel 28. The ink is collected to the outside of the liquid ejection substrate 2 through these connection channels 15. The second common supply channel 27 and the second common collection channel 28 extend in the longitudinal direction of the liquid ejection substrate 2. Also, a plurality of second common supply channels 27 and a plurality of second common collection channels 28 are formed alternately in the transverse direction of the liquid ejection substrate 2. As illustrated in FIGS. 3A and 3B, each first common supply channel 17 and the corresponding second common supply channel 27 together form a common supply channel. Likewise, each first common collection channel 18 and the corresponding second common collection channel 28 together form a common collection channel.

The ejection port formation substrate 201, the vibration substrate 202, the liquid supply substrate 203, the channel formation substrate 204, and the damper substrate 302 can each be a silicon substrate or the like. Also, an example in which the substrates are separate substrates has been described in the present embodiment, but they are not limited to separate ones. The damper member 300 is made of an elastic material. For example, resin materials such as polyimides and polyamides are usable. The method of forming openings in the damper member 300 includes dry etching. Patterning using light exposure may be employed in a case where the damper member is a photosensitive resin.

As described above, each liquid ejection substrate 2 has: a first substrate having the ejection ports 3 formed therein (ejection port formation substrate 201); a second substrate having the pressure chambers 5 formed therein (vibration substrate 202); and a third substrate having the individual supply channels 7 and the individual collection channels 8 formed therein (liquid supply substrate 203). The liquid ejection substrate 2 further has: a fourth substrate including the damper member 300 and having the first common supply channels 17 and the first common collection channels 18 formed therein (damper substrate 302); and a fifth substrate having the second common supply channels 27 and the second common collection channels 28 formed therein (channel formation substrate 204). Moreover, the first sub-

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strate (ejection port formation substrate 201), the second substrate (vibration substrate 202), the third substrate (liquid supply substrate 203), the fourth substrate (damper substrate 302), and the fifth substrate (channel formation substrate 204) are laminated in this order.

The channel formation substrate 204 has a first surface to be laminated to the damper substrate 302 and a second surface opposite to the first surface. Moreover, the channel formation substrate 204 has through-holes penetrating through the first surface and the second surface (the portions of the connection channels 15). Furthermore, recesses that function as the damper areas 301 are formed in the first surface of the channel formation substrate 204. The through-holes and the recesses are disposed alternately in the transverse direction of the liquid ejection substrate 2 (Y direction).

<Arrangement of Ejection Ports and Arrangement of Dampers>

FIG. 4 is a plan view explaining channel portions in a liquid ejection substrate 2 in the present embodiment. FIG. 4 illustrates a part of the liquid ejection substrate 2. The longitudinal direction of the liquid ejection substrate 2 is the left-right direction in the plane of the drawing sheet (X direction). The transverse direction of the liquid ejection substrate 2 is the up-down direction in the plane of the drawing sheet (Y direction). As illustrated in FIG. 4, a plurality of ejection ports 3 are disposed along the longitudinal direction of the liquid ejection substrate 2, which is the X direction, and form an ejection port array. A plurality of ejection port arrays thus formed are provided in the transverse direction of the liquid ejection substrate 2 (Y direction).

FIG. 5 is a view illustrating a cross section around ejection ports 3 in the present embodiment. FIG. 5 is a view illustrating a cross section indicated by the V-V line in FIG. 4. As illustrated in FIG. 5, channel partitions 16, which are formed by the damper substrate 302, are provided between the first common supply channels 17 and the first common collection channels 18 in the damper substrate 302. The channel partitions 16 of the damper substrate 302 are affixed to the liquid supply substrate 203 with a bonding layer 19.

As illustrated in FIGS. 3A, 3B, and 4, the second common supply channels 27 and the second common collection channels 28 are formed so as to extend in the direction along the ejection port arrays (i.e., the longitudinal direction of the liquid ejection substrate 2). The individual supply channels 7 communicating with the pressure chambers 5 each communicate with the corresponding second common supply channel 27 through the corresponding first common supply channel 17. The individual collection channels 8 communicating with the pressure chambers 5 each communicate with the corresponding second common collection channel 28 through the corresponding first common collection channel 18. The second common supply channels 27 and the second common collection channels 28 are formed so as to extend in the direction along the ejection port arrays. This allows the pressure chambers 5 corresponding to the respective ejection ports 3 to form ejection port arrays in such a manner as to be adjacent to one another in their transverse direction. Accordingly, in each liquid ejection substrate 2 in the present embodiment, the ejection ports 3 are densely disposed.

For example, in FIG. 4, the length of the pressure chambers 5 in their transverse direction (X direction in FIG. 4) is 110 μm , and the pressure chambers 5 and the ejection ports 3 are disposed at intervals of 150 dpi in the form of ejection port arrays. Four of such ejection port arrays are arranged so

as to be offset from one another in the longitudinal direction of the pressure chamber **5** (the Y direction in FIG. 4) and to be offset from one another in the transverse direction of the pressure chamber **5** (X direction in FIG. 4). This arrangement enables a high ejection port density of 600 dpi on a print medium. In the present embodiment, four ejection port arrays are disposed to achieve 600 dpi. Alternatively, the configuration may be such that eight ejection port arrays are disposed to achieve 1200 dpi.

In the case where the ejection ports **3** are disposed thus densely, crosstalk may occur in which a pressure fluctuation occurring in each pressure chamber **5** propagates to other pressure chambers **5** and affects ejection characteristics. To address this, in the present embodiment, dampers are provided on walls of the first common supply channels **17** and the first common collection channels **18** extending in the direction along the ejection port arrays, which is the X direction. Specifically, the damper areas **301** are provided on walls of the first common supply channels **17** and the first common collection channels **18** extending in the longitudinal direction of the liquid ejection substrate **2**, the walls extending in the longitudinal direction. In this way, the damper areas **301** are large as compared to a case where the damper areas are provided in the transverse direction of the substrate, and therefore absorb pressures sufficiently.

A pressure generated in each pressure chamber **5** propagates through the individual supply channel **7** and the individual collection channel **8** to other pressure chambers **5**. For this reason, the damper areas **301** are provided on the walls of the first common supply channels **17** and the first common collection channels **18** at positions opposed to the individual supply channels **7** and the individual collection channels **8**. As illustrated in FIG. 4, in the channel formation substrate **204**, the damper areas **301** and the second common supply channels **27** and the second common collection channels **28** are disposed adjacently to each other in the direction in which the ejection port arrays are disposed side by side (Y direction). In the example illustrated in FIG. 4, the channels and the damper areas are disposed alternately in the order of a second common collection channel **28**, a damper area **301**, a second common supply channel **27**, a damper area **301**, and so on from the near side in the Y direction. Assume that the interval between ejection port arrays is approximately 1 mm in the example of FIG. 4. In this case, the interval between each damper area and each adjacent common channel can be 0.1 mm by setting the length of the damper area **301** in the Y direction in FIG. 4 at 0.5 mm and setting the lengths of the second common supply channels **27** and the second common collection channels **28** in the Y direction at 0.3 mm. That is, damper areas **301** of a sufficiently large size can be provided even in a case where the ejection ports **3** and the pressure chambers **5** are disposed densely. In FIG. 4, the length of the ejection port arrays determines the length of the damper areas **301** in their longitudinal direction (X direction). In the present embodiment, the damper areas **301** are provided to be longer than the ejection port arrays. This ensures a crosstalk suppression effect up to the ejection ports **3** at the ends of the ejection port arrays.

In the example described above, the first common supply channels **17** and the first common collection channels **18** are provided with the damper areas **301**. However, the present embodiment is not limited to this example. FIG. 8 is a diagram illustrating a modification of the present embodiment. In the present embodiment, as illustrated in FIG. 8, only the first common collection channels **18** may be provided with the damper areas **301**, and the first common

supply channels **17** may be formed narrower without the damper areas **301**. Providing a damper area **301** at least at one location brings about a pressure absorption effect. By selecting the channels to provide the damper areas **301** with as described above, nozzles can be disposed more densely. Moreover, narrowing the first common supply channels **17** without the damper areas **301** is advantageous since it reduces the size of the substrates in the transverse direction (Y direction) and thus lowers the substrate cost. On the other hand, the larger the number of damper areas, the higher the pressure absorption effect. It is therefore preferable to dispose the damper areas at all of the first common supply channels **17** and the first common collection channels **18** as illustrated in FIGS. 3A to 5. In FIGS. 3A, 3B, and 4, an example has been described in which a plurality of first common supply channels **17**, a plurality of second common supply channels **27**, a plurality of first common collection channels **18**, and a plurality of second common collection channels **28** are provided. However, it suffices that at least one of each is provided.

As described above, according to the present embodiment, it is possible to suppress crosstalk while disposing the ejection ports **3** densely. It is therefore possible to stabilize the ejection characteristics and obtain images with a higher image quality and a higher definition.

Second Embodiment

In the first embodiment, an example has been described in which the damper substrate **302** is included, and the first common supply channels **17** and the first common collection channels **18** are formed in the damper substrate **302**. In a second embodiment, an example in which the first common supply channels **17** and the first common collection channels **18** are formed in the liquid supply substrate **203** will be described.

FIG. 6 is a view illustrating a cross section around ejection ports **3** in the present embodiment. Like FIG. 5, FIG. 6 is a view illustrating across section indicated by the V-V line in FIG. 4. As illustrated in FIG. 6, in the present embodiment, the individual supply channels **7** communicate with the first common supply channels **17** formed in the liquid supply substrate **203**. The individual collection channels **8** communicate with the first common collection channels **18** formed in the liquid supply substrate **203**.

Moreover, in the present embodiment, the damper member **300** is formed on the channel formation substrate **204**. Furthermore, the damper member **300** forms the walls of the first common supply channels **17** formed in the liquid supply substrate **203** which face the individual supply channels **7**, and the walls of the first common collection channels **18** formed in the liquid supply substrate **203** which face the individual collection channels **8**. In the present embodiment, the damper substrate **302** as described in the first embodiment is omitted by providing the damper member **300** on the channel formation substrate **204**.

As described above, each liquid ejection substrate **2** in the present embodiment has a first substrate having the ejection ports **3** formed therein (ejection port formation substrate **201**) and a second substrate having the pressure chambers **5** formed therein (vibration substrate **202**). The liquid ejection substrate **2** further has a third substrate having the individual supply channels **7**, the individual collection channels **8**, the first common supply channels **17**, and the first common collection channels **18** formed therein (liquid supply substrate **203**). The liquid ejection substrate **2** further has a fourth substrate having the second common supply channels

27 and the second common collection channels 28 (channel formation substrate 204). Moreover, the first substrate (ejection port formation substrate 201), the second substrate (vibration substrate 202), the third substrate (liquid supply substrate 203), and the fourth substrate (channel formation substrate 204) are laminated in this order.

The liquid ejection substrate 2 is formed by affixing the substrate having the damper member 300. In the first embodiment, the damper substrate 302 having the damper member 300 is affixed to the liquid supply substrate 203 with the bonding layer 19. In the present embodiment, on the other hand, the channel formation substrate 204 having the damper member 300 is affixed to the liquid supply substrate 203. According to the present embodiment, it is possible to reduce costs and enhance the degree of freedom in design. A description will be given below while comparing with an example of the first embodiment.

In the example of the first embodiment illustrated in FIG. 5, a distance D represents the distance between an opening of an individual supply channel 7 and the bonding layer 19. The distance D is required to be such a sufficient length that the bonding layer 19, if sticking out, will not close the opening of the individual supply channel 7. The first common supply channels 17 and the first common collection channels 18 are therefore required to be designed with the bonding layer 19 and its sticking area taken into consideration. On the other hand, forming the first common supply channels 17 and the first common collection channels 18 in the liquid supply substrate 203 as in the present embodiment illustrated in FIG. 6 eliminates the possibility of the bonding layer 19 closing openings of the individual supply channels 7 and the individual collection channels 8. This enables each individual channel and each common channel to be formed with a desired design. Also, since the damper substrate 302 is omitted, the number of substrates to be bonded in the forming of the liquid ejection substrate 2 is reduced. Reducing the number of substrates reduces costs, reduces the bonding cost required for bonding of the substrates, and, as mentioned earlier, enhances the degree of freedom in design. Moreover, the nozzles can be disposed more densely by selecting the channels to provide the damper areas 301 with as described in the first embodiment. FIG. 9 is a view illustrating a configuration in the second embodiment in which only the first common collection channels 18 are provided with the damper areas 301, and the first common supply channels 17 are formed narrower without the damper areas 301.

FIG. 7 is a view illustrating a modification of the present embodiment. FIG. 7 is a view illustrating a cross section around ejection ports 3, and is a view illustrating a cross section indicated by the V-V line in FIG. 4. As illustrated in FIG. 7, patterns in which minute holes are formed can be formed at areas of the damper member 300 between the second common supply channels 27 and the first common supply channels 17. In this way, the areas of the damper member 300 where the patterns are formed will function as filters. The filters may be formed only on the supply side as in the example of FIG. 7. Alternatively, the filters formed of the damper member 300 may be formed also between the first common collection channels 18 and the second common collection channels 28 on the discharge side. The modification illustrated in FIG. 7 is not limited to the second embodiment. The modification is applicable also to a case of using the damper substrate 302 to form the damper areas 301 as described in the first embodiment. Specifically, in the configuration illustrated in FIG. 5, patterns may be formed at the portions of the damper member 300 between the

second common supply channels 27 and the first common supply channels 17 to impart a filtering function. Similarly, in the configuration illustrated in FIG. 5, patterns may be formed at the portions of the damper member 300 between the second common collection channels 28 and the first common collection channels 18 to impart a filtering function.

Third Embodiment

In the first embodiment, an example has been described in which the channel partitions 16, which separate the first common supply channels 17 and the first common collection channels 18 from each other with the bonding layer 19, are bonded to the damper substrate 302. In a third embodiment, an example in which the bonding layer 19 is not provided on the channel partitions 16 will be described.

FIGS. 10A and 10B are views illustrating cross sections around ejection ports 3 in the present embodiment. FIGS. 10A and 10B are views along cross-sectional lines set through connection channels 15. FIG. 10A represents an example in which the first common supply channels 17 and the first common collection channels 18 are provided with the damper areas 301. FIG. 10B represents an example in which only the first common collection channels 18 are provided with the damper areas 301, and the first common supply channels 17 are formed narrower without the damper areas 301.

As illustrated in FIGS. 10A and 10B, the bonding layer 19 is not provided on the channel partitions 16 between the first common supply channels 17 and the first common collection channels 18, and a minute communication portion 20 is provided there. Unlike a case of using a usual bonding layer 19 to affix the substrates, this configuration makes it possible to reduce the size of the areas of the channel partitions 16. This in turn makes it possible to increase the sizes of the areas of the damper areas 301, the first common supply channels 17, and the first common collection channels 18. Accordingly, the damper areas 301 can be formed wider, which will further enhance the pressure absorption effect.

Also, part of the ink flows into the first common collection channels 18 from the first common supply channels 17 through the minute communication portion 20. This brings about a further effect in which the minute communication portion 20 is connected so as to reduce stagnation at stagnating regions on the damper areas 301 where circulatory flows 21 do not easily flow. This facilitates the flow of bubbles and so on in the first common supply channels 17 and the first common collection channels 18 by the circulatory flows 21.

Incidentally, in a case where the dimension of the minute communication portion 20 is larger than a predetermined value, the circulatory flows flowing through the minute communication portion 20 will be so large that the circulatory flows 21 flowing through the individual supply channels 7, the pressure chambers 5, and the individual collection channels 8 in this order will be small. For this reason, the dimension of the minute communication portion 20 is preferably small, and the channel resistance of the minute communication portion 20 is preferably small. FIG. 11 is graph illustrating the height of the minute communication portion versus a resistance ratio. In FIG. 11, the horizontal axis represents the height of the minute communication portion 20, and the vertical axis represents the ratio between the viscous resistance of the minute communication portion 20 and the viscous resistance of ejection port channels (channels from the individual supply channels 7 through the

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pressure chambers **5** to the individual collection channels **8**). The resistance ratio represents the ratio between the flow rate of the ink flowing through the minute communication portion **20** and that of the ejection port channels. The viscous resistance of the channel at the minute communication portion **20** is 100 times the viscous resistance of the ejection port channels or more and desirably 1000 times or more. The height of the minute communication portion **20** in the direction of lamination of the substrates is 7 μm or less and desirably 3 μm or less.

The minute communication portion **20** described above is similarly usable in the example described in the second embodiment. In the second embodiment, the channel partitions **16**, which separate the first common supply channels **17** and the first common collection channels **18** from each other with the bonding layer **19**, are bonded to the channel formation substrate **204**. FIGS. **12A** and **12B** are views illustrating cross sections around ejection ports in a case where the present embodiment is applied to the configuration described in the second embodiment. Like FIGS. **10A** and **10B**, FIGS. **12A** and **12B** are views along cross-sectional lines set through connection channels **15**. FIG. **12A** represents an example in which the first common supply channels **17** and the first common collection channels **18** are provided with the damper areas **301**. FIG. **12B** represents an example in which only the first common collection channels **18** are provided with the damper areas **301**, and the first common supply channels **17** are formed narrower without the damper areas **301**. A configuration in which the bonding layer **19** is not provided on the channel partitions **16** between the first common supply channels **17** and the first common collection channels **18**, and the minute communication portion **20** is provided there, as illustrated in FIGS. **12A** and **12B**, is used. In this case too, advantageous effects similar to those described earlier can be achieved.

As described above, according to the present embodiment, the damper areas **301** can be formed wider. This further enhances the pressure absorption effect. The stagnation at the stagnating regions on the damper areas **301** where the circulatory flows **21** do not easily flow can be reduced.

OTHER EMBODIMENTS

In the above embodiments, piezoelectric elements have been exemplarily described as the pressure generating elements that generate a pressure in the pressure chambers. Any elements may be used as the pressure generating elements. For example, heating elements that generate a pressure by generating a bubble by heating may be used.

While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

This application claims the benefit of Japanese Patent Application No. 2022-056336, filed Mar. 30, 2022, and No. 2022-082418, filed May 19, 2022, which are hereby incorporated by reference herein in their entirety.

What is claimed is:

1. A liquid ejection head comprising:

a plurality of ejection ports configured to eject a liquid, the ejection ports being disposed in an ejection port array;

a plurality of pressure chambers corresponding respectively to the plurality of ejection ports and communicating with the ejection ports;

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a plurality of individual supply channels corresponding respectively to the plurality of pressure chambers and communicating with the pressure chambers;

a plurality of individual collection channels corresponding respectively to the plurality of pressure chambers and communicating with the pressure chambers;

a common supply channel communicating with the plurality of individual supply channels, the common supply channel communicating with surfaces of the individual supply channels opposite to surfaces thereof communicating with the pressure chambers;

a common collection channel communicating with the plurality of individual collection channels, the common collection channel communicating with surfaces of the individual collection channels opposite to surfaces thereof communicating with the pressure chambers;

a damper member forming a wall of a part of at least one of the common supply channel or the common collection channel;

a first substrate having the ejection ports formed therein; a second substrate having the pressure chambers formed therein;

a third substrate having the individual supply channels and the individual collection channels formed therein;

a fourth substrate including the damper member and having the common supply channel and the common collection channel formed therein; and

a fifth substrate having a second common supply channel and a second common collection channel formed therein, the second common supply channel communicating with the common supply channel, and the second common collection channel communicating with the common collection channel, wherein

the common supply channel and the common collection channel are formed so as to extend in a first direction along the ejection port array,

the common supply channel and the common collection channel are disposed side by side in a second direction crossing the ejection port array, and

the first substrate, the second substrate, the third substrate, the fourth substrate, and the fifth substrate are laminated in the order as listed.

2. The liquid ejection head according to claim 1, wherein the damper member forms the wall of the part of one of the common supply channel or the common collection channel.

3. The liquid ejection head according to claim 1, wherein the damper member forms the walls of the parts of the common supply channel and the common collection channel.

4. The liquid ejection head according to claim 1, wherein a plurality of the ejection port arrays are formed side by side in the second direction, and

a plurality of the common supply channels and a plurality of the common collection channels are provided and are disposed alternately in the second direction.

5. The liquid ejection head according to claim 1, wherein the individual supply channels and the individual collection channels are formed so as to extend in a direction crossing the first direction and the second direction.

6. The liquid ejection head according to claim 1, wherein a length of the ejection port array is shorter than a length of the damper member in the first direction.

7. The liquid ejection head according to claim 1, wherein the fifth substrate includes:

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a through-hole penetrating through a first surface to be laminated to the fourth substrate and a second surface being an opposite surface to the first surface, and

a recess formed in the first surface, and
the through-hole and the recess are disposed side by side in the second direction.

8. The liquid ejection head according to claim 1, further comprising a bonding layer provided between the third substrate and the fourth substrate.

9. The liquid ejection head according to claim 8, wherein the bonding layer is formed so as to provide a minute communication portion on a channel partition between the common supply channel and the common collection channel.

10. The liquid ejection head according to claim 9, wherein viscous resistance of the minute communication portion is at least 100 times viscous resistance of channels from the individual supply channels through the pressure chambers to the individual collection channels.

11. The liquid ejection head according to claim 9, wherein a height of the minute communication portion in a direction of lamination of the substrates is 7 μm or less.

12. The liquid ejection head according to claim 8, wherein the bonding layer is provided between the third substrate and a channel partition separating the common supply channel and the common collection channel in the fourth substrate.

13. A liquid ejection head comprising:

a plurality of ejection ports configured to eject a liquid, the ejection ports being disposed in an ejection port array;

a plurality of pressure chambers corresponding respectively to the plurality of ejection ports and communicating with the ejection ports;

a plurality of individual supply channels corresponding respectively to the plurality of pressure chambers and communicating with the pressure chambers;

a plurality of individual collection channels corresponding respectively to the plurality of pressure chambers and communicating with the pressure chambers;

a common supply channel communicating with the plurality of individual supply channels, the common supply channel communicating with surfaces of the individual supply channels opposite to surfaces thereof communicating with the pressure chambers;

a common collection channel communicating with the plurality of individual collection channels, the common collection channel communicating with surfaces of the individual collection channels opposite to surfaces thereof communicating with the pressure chambers;

a damper member forming a wall of a part of at least one of the common supply channel or the common collection channel;

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a first substrate having the ejection ports formed therein; a second substrate having the pressure chambers formed therein;

a third substrate having the individual supply channels, the individual collection channels, the common supply channel, and the common collection channel formed therein; and

a fourth substrate including the damper member and having a second common supply channel and a second common collection channel formed therein, the second common supply channel communicating with the common supply channel, and the second common collection channel communicating with the common collection channel, wherein

the common supply channel and the common collection channel are formed so as to extend in a first direction along the ejection port array,

the common supply channel and the common collection channel are disposed side by side in a second direction crossing the ejection port array, and

the first substrate, the second substrate, the third substrate, and the fourth substrate are laminated in the order as listed.

14. The liquid ejection head according to claim 13, wherein

the fourth substrate includes:

a through-hole penetrating through a first surface to be laminated to the third substrate and a second surface opposite to the first surface, and

a recess formed in the first surface, and the through-hole and the recess are disposed side by side in the second direction.

15. The liquid ejection head according to claim 14, wherein the fourth substrate includes the damper member on the first surface to be laminated to the third substrate.

16. The liquid ejection head according to claim 13, further comprising a bonding layer provided between the third substrate and the fourth substrate.

17. The liquid ejection head according to claim 16, wherein the bonding layer is formed so as to provide a minute communication portion on a channel partition between the common supply channel and the common collection channel.

18. The liquid ejection head according to claim 17, wherein viscous resistance of the minute communication portion is at least 100 times viscous resistance of channels from the individual supply channels through the pressure chambers to the individual collection channels.

19. The liquid ejection head according to claim 17, wherein a height of the minute communication portion in a direction of lamination of the substrates is 7 μm or less.

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